

Reasoning for Code Changes

Certainly! Here are the clear, concise, and structured reasons behind the changes made in the unified diff:

- **Updated javax to jakarta for Spring Boot 3 Compatibility**
- **Files Affected:** `User.java`, `RegisterRequestDto.java`, `Book.java`, `Tag.java`, `BorrowRecord.java`, `GlobalExceptionHandler.java`, `JwtRequestFilter.java`, `BookDto.java`, `GlobalExceptionHandler.java` (multiple instances)

Reason: Spring Boot 3 has transitioned from using `javax` packages to `jakarta` packages to align with the Jakarta EE 9+ standards. This change is necessary to ensure compatibility with the latest version of Spring Boot and to avoid runtime errors due to missing or deprecated classes.

Fixed Hibernate 6 Compatibility Issue in `BorrowRecord.java`

- **Files Affected:** `BorrowRecord.java`

Reason: Hibernate 6 no longer supports the `@Type` annotation for custom types like `yes_no`. The `@Type` annotation has been replaced with `@Convert(converter = YesNoConverter.class)` to ensure compatibility with Hibernate 6 and to avoid compilation errors.

Corrected and Simplified Method Signatures in `GlobalExceptionHandler.java`

- **Files Affected:** `GlobalExceptionHandler.java` (multiple instances)

Reason: The repeated method signatures for `handleBookNotFound` and the general exception handler were causing syntax errors. The changes ensure that each method is defined only once, maintaining the correct structure and avoiding compile-time errors.

Ensured `createdAt` Field is Not Updatable in `BorrowRecord.java`

- **Files Affected:** `BorrowRecord.java`

Reason: The `createdAt` field was marked with `@PrePersist` to set the creation timestamp, but it was missing the `updatable = false` attribute. This change ensures that the `createdAt` field is only set once when the entity is created and cannot be updated later, maintaining data integrity.

Formatted and Cleaned Up Code in `BorrowRecord.java` and `GlobalExceptionHandler.java`

- **Files Affected:** `BorrowRecord.java`, `GlobalExceptionHandler.java` (multiple instances)

Reason: The code was reformatted to improve readability and maintain consistency. This includes proper indentation, removing redundant `private` keyword repetitions, and ensuring method signatures are correctly defined. These changes make the code easier to read and maintain.

Fixed Redundant @Column Annotations in BorrowRecord.java

- **Files Affected:** BorrowRecord.java

Reason: There were redundant @Column annotations for the status and fineAmount fields. These annotations were cleaned up to avoid confusion and ensure that the annotations are used correctly and consistently.

Ensured Proper Return Statement in GlobalExceptionHandler.java

- **Files Affected:** GlobalExceptionHandler.java (multiple instances)
- **Reason:** The return statement in the general exception handler was incorrectly split across multiple lines, causing syntax errors. The changes ensure that the return statement is correctly formatted and the method returns the expected response.

These changes collectively ensure that the codebase is compatible with Spring Boot 3, Hibernate 6, and modern Java standards, while also improving readability and maintainability.